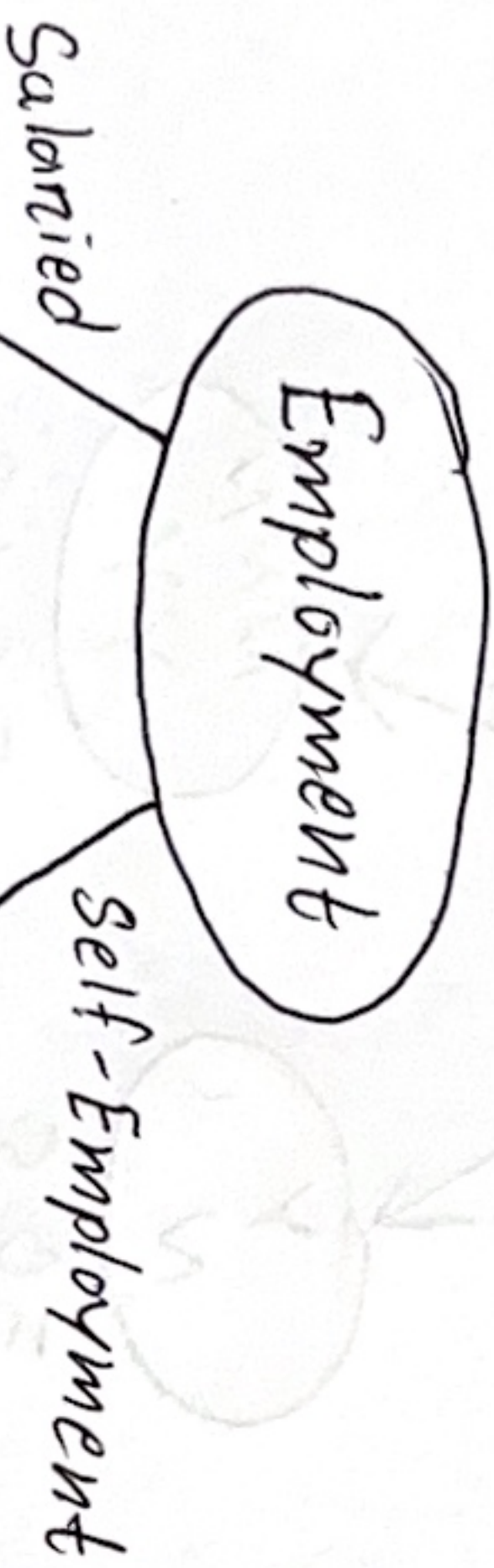


CT-2 (Gini Impurity)

ID	Employment	Credit-History	Dependents	Loan-Approved
1	Salaries	Good	3	Yes
2	Salaries	Bad	1	No
3	Self-Employment	Good	4	Yes
4	Self-Employment	Bad	2	No
5	Salaries	Good	0	Yes
6	Self-Employment	Good	5	No
7	Salaries	Bad	2	No

Considering

Employment



$$Gini = 1 - \sum p_i^2$$

$$Gini = 1 - \sum_{i=1}^n (p_i)^2$$

$$\text{Weighted Gini} = \sum_{i=1}^k \frac{n_i}{N} Gini(i)$$

$$Gini = 1 - \left(\frac{2}{4}\right)^2 - \left(\frac{2}{4}\right)^2$$

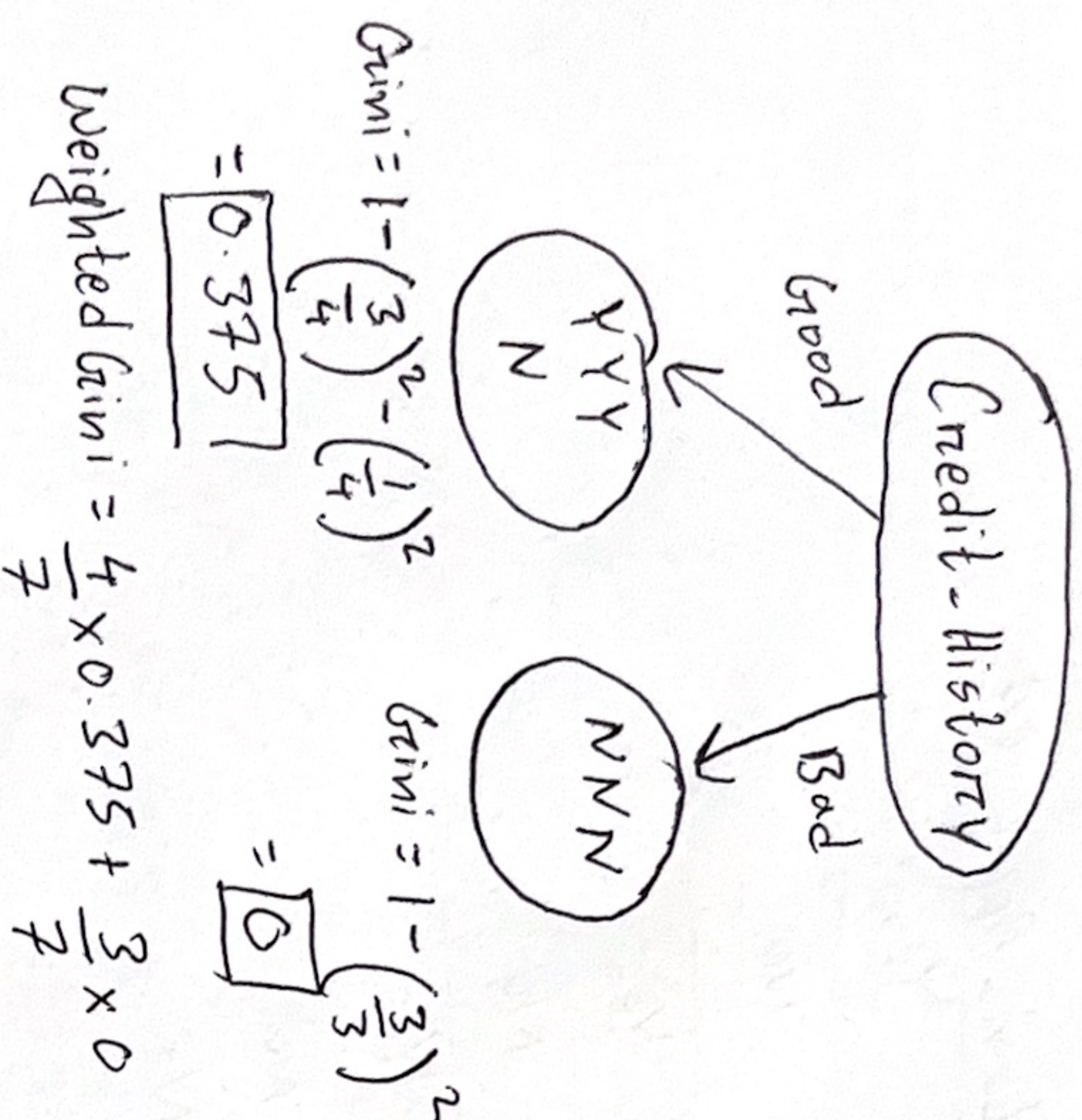
$$= 0.5$$

$$Gini = 1 - \left(\frac{1}{3}\right)^2 - \left(\frac{2}{3}\right)^2$$

$$= 0.444$$

$$\text{Weighted Gini} = \frac{4}{7} \times 0.5 + \frac{3}{7} \times 0.444 = 0.476$$

Considering Credit History

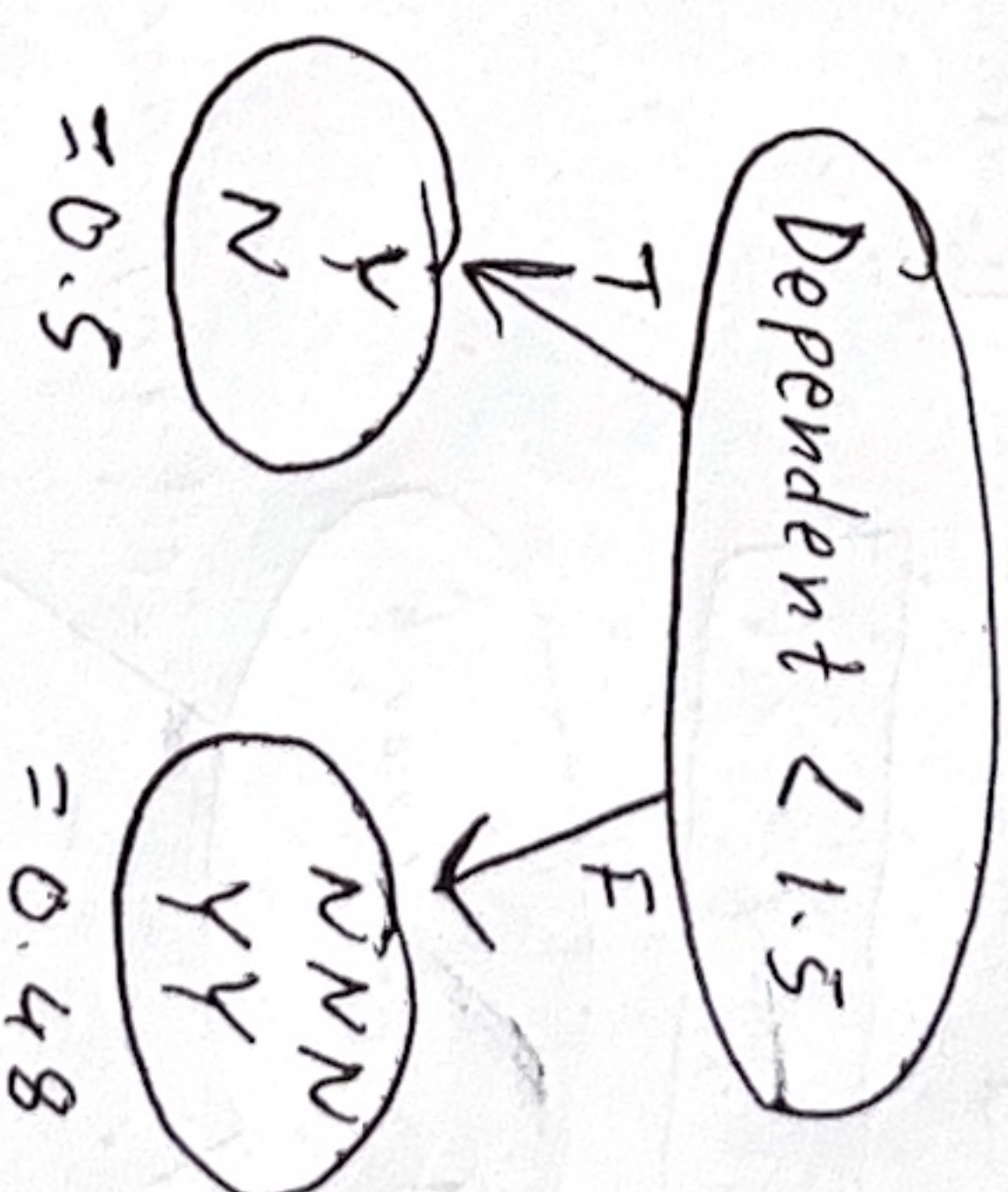
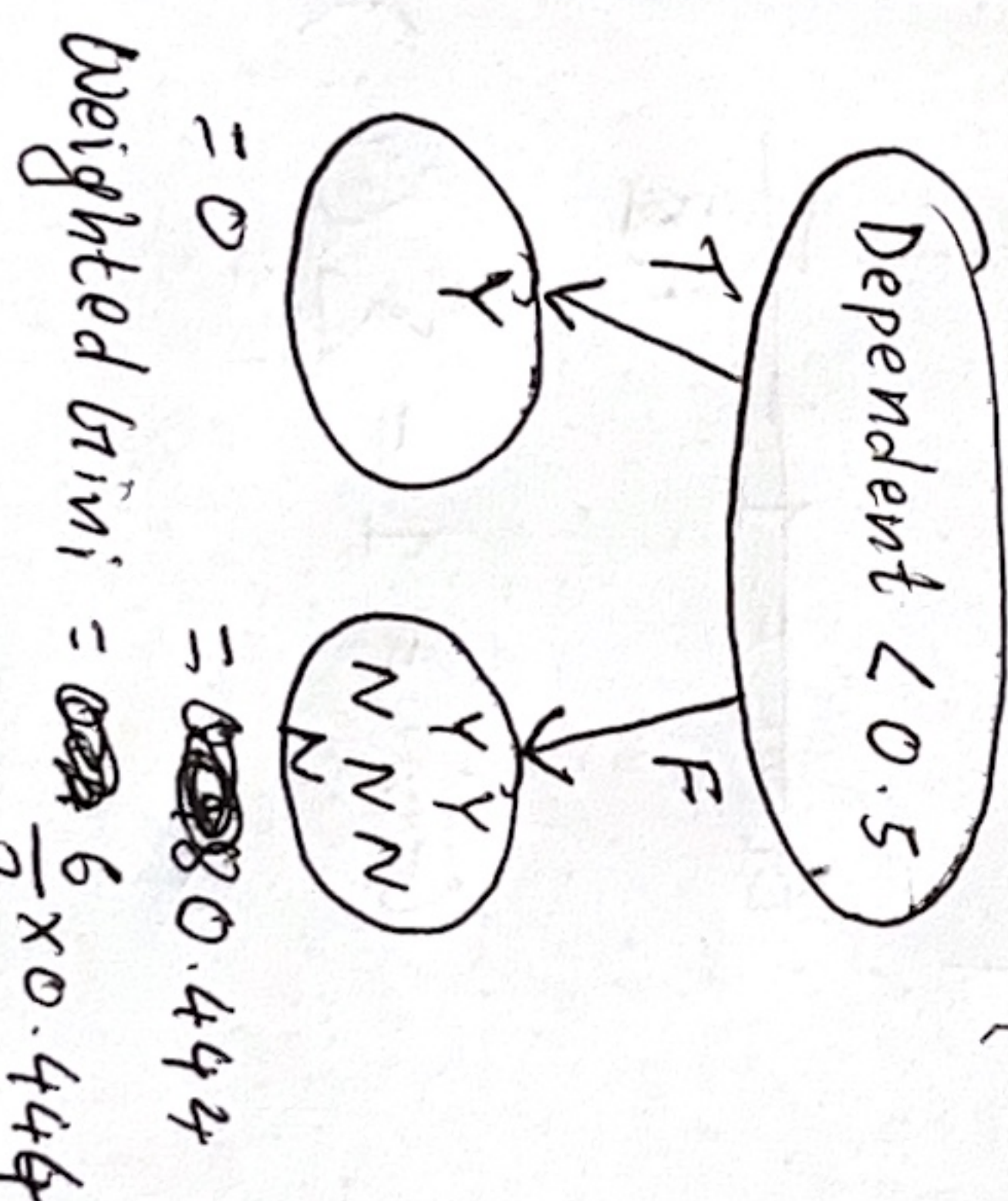


Considering Dependents

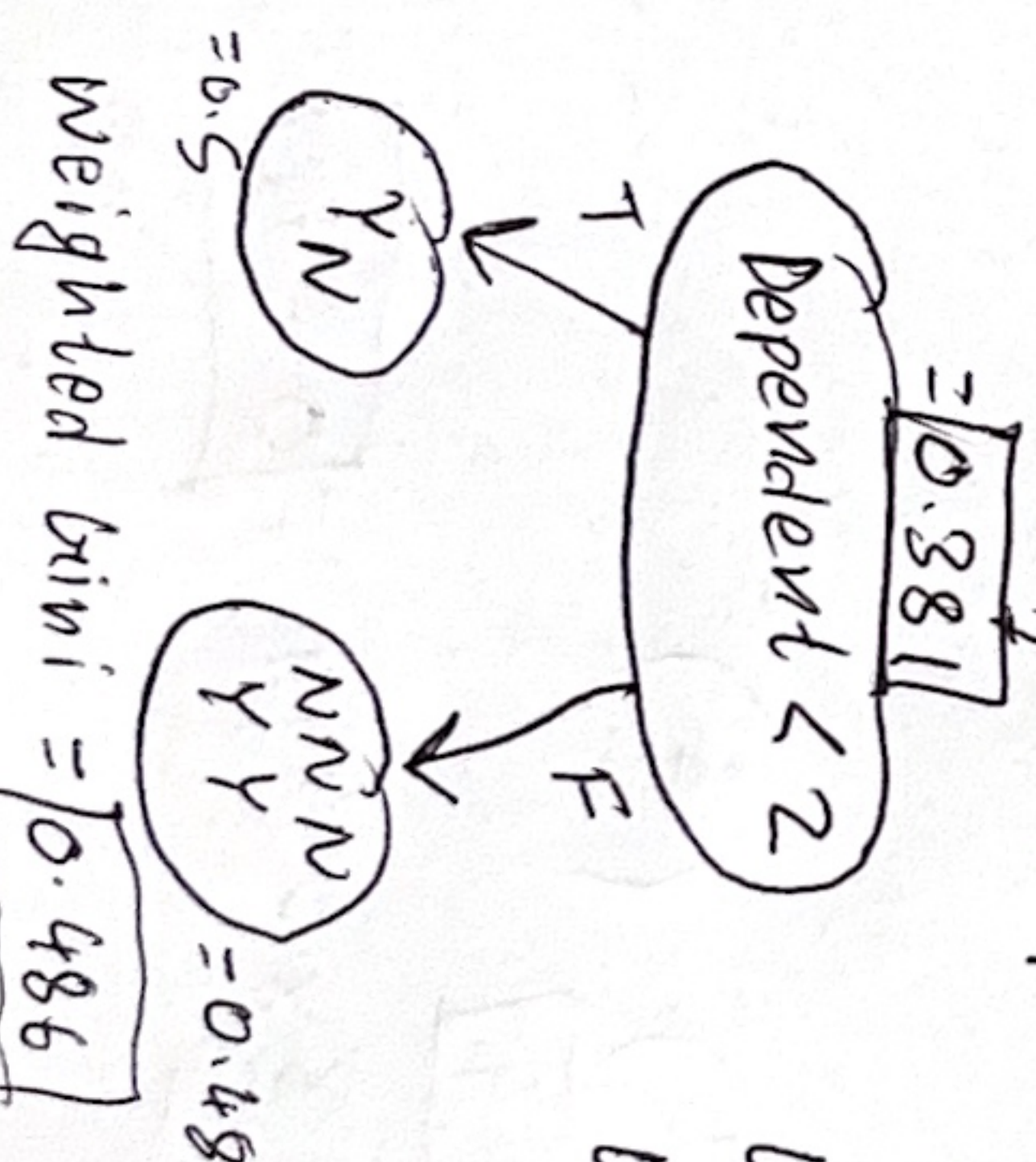
Sorting table respect to Dependent

ID	Dependents	Loan-Approved
5	0	Yes
2	1	No
4	2	No
7	2	No
1	3	Yes
3	4	Yes
6	5	No

0 - 0.5
1 - 1.5
2 - 2
3 - 2.5
4 - 3.5
5 - 4.5

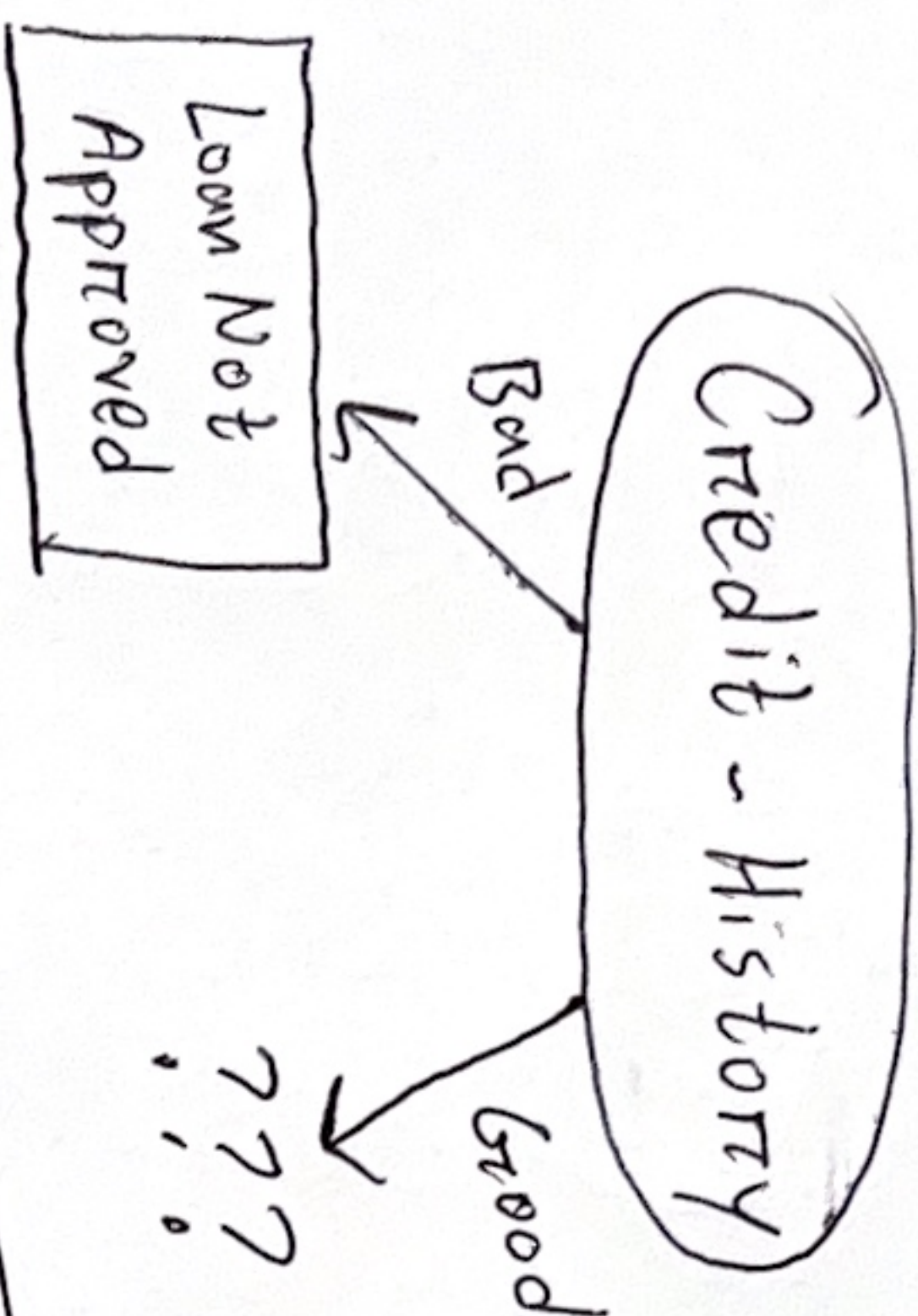


Weighted Gini (< 2.5) = 0.404
 Weighted Gini (< 3.5) = 0.486
 Weighted Gini (< 4.5) = 0.429



Comparing All Weighted Gini
We found credit-history has
the lowest value $[0.214]$

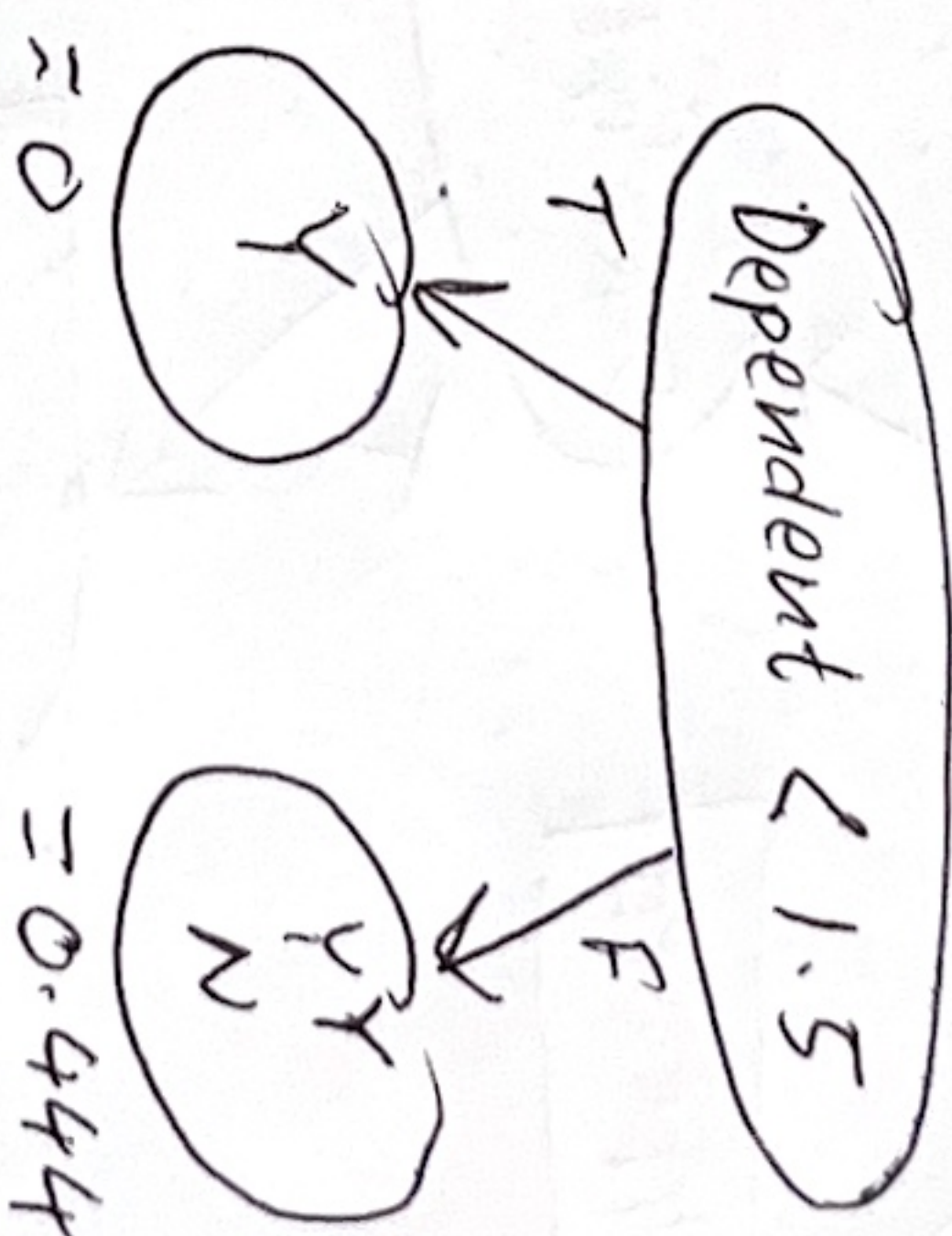
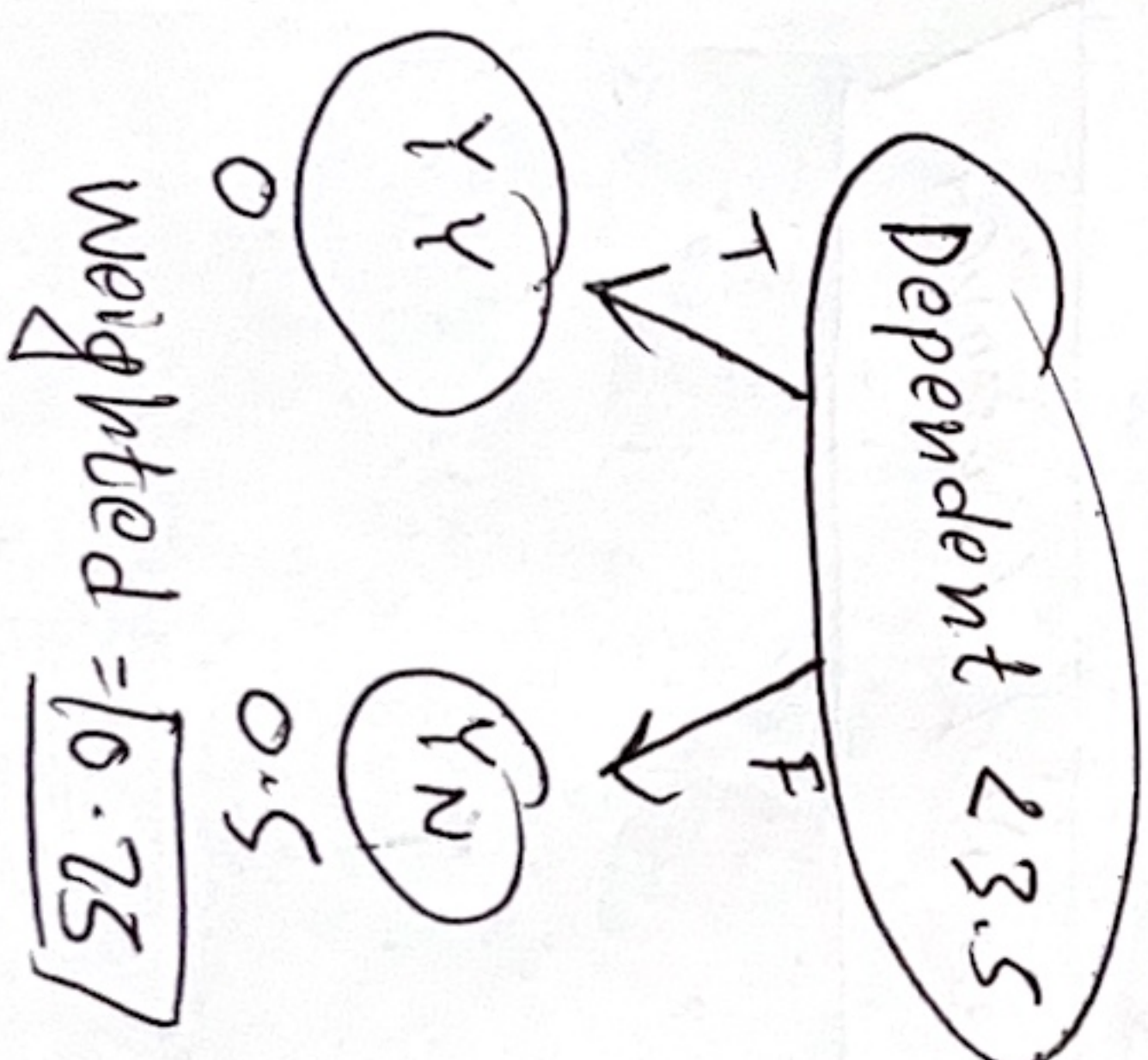
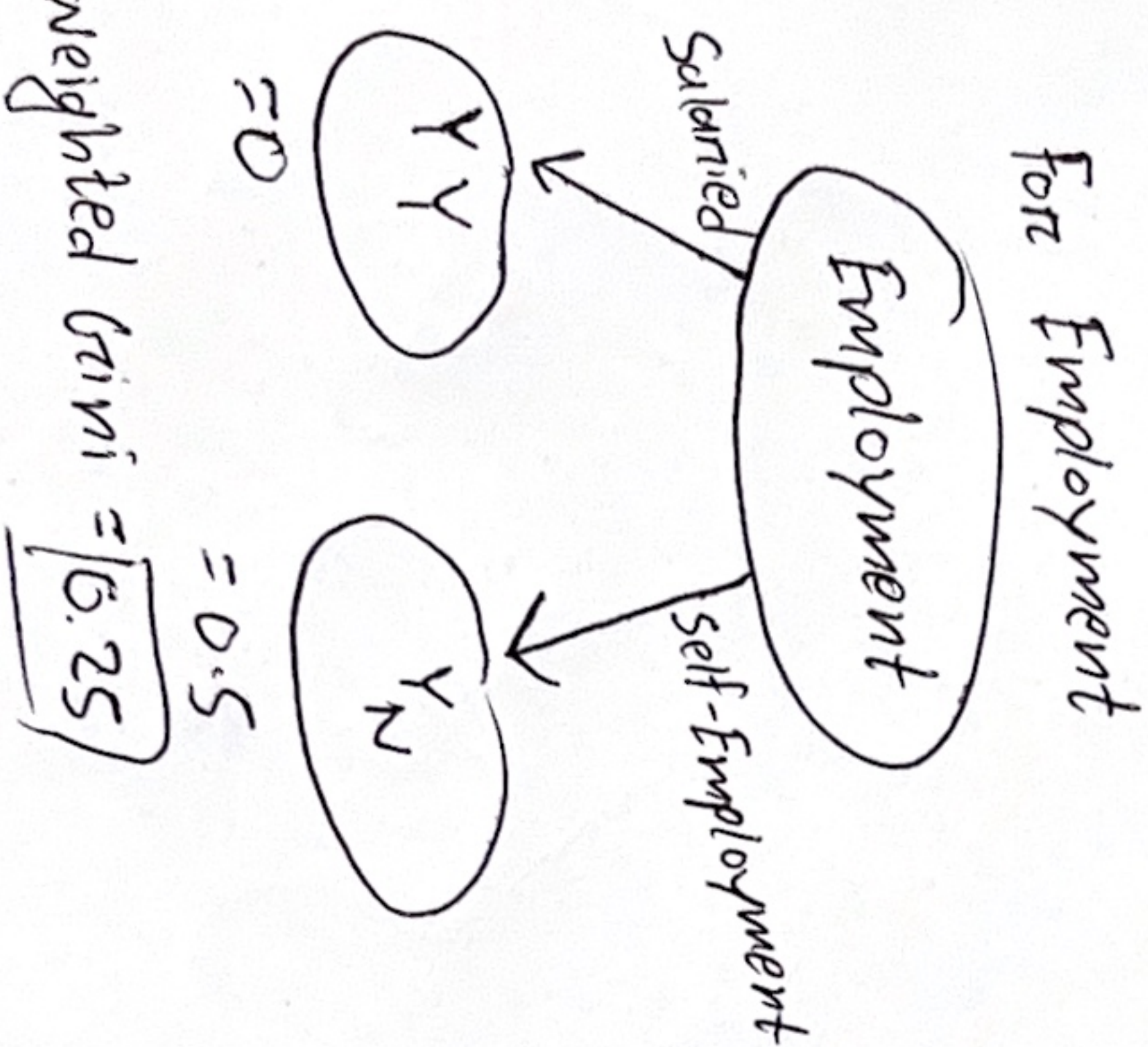
$\therefore$ Root Node



ID	Employment	Credit-History	Dependent	Loan-Approved
1	Salaried	Good	3	Yes
3	Self-Employment	Good	4	Yes
5	Salaried	Good	0	Yes
6	Self-Employment	Good	5	No

For Dependent

Q	Yes
1.5 - 3	Yes
3.5 - 4	Yes
4.5 - 5	No



$$\text{Weighted} = \frac{3}{4} \times 0.4444$$

$$= [0.333]$$



Among all weighted Gini
Dependent < 4.5 is the
lowest ::

